# Xiongyule Zhou

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China

## EDUCATION

**Beijing Jiaotong University**, Beijing, China Sep 2023 – Jul 2027 (Expected)

Bachelor of Engineering in Mechatronics Engineering

Cumulative GPA: **3.67/4.0** (88.2/100)

Relevant Coursework: Artificial Intelligence, Robotics and Flexible Automation, Control Theory, Embedded Systems, Signals and Systems, Intelligent Mechatronic System Design

## RESEARCH EXPERIENCE

**Learning-Based Robotic Manipulation**, Beijing Jiaotong University Sep 2025 – Feb 2026

- Developed a Python/PyTorch learning pipeline for a simulated robotic arm, using MATLAB/Simulink to model robot dynamics for grasping, target placement and object-handoff tasks.
- Applied DINOv2 and OpenCV-based visual representations to extract object-level features for vision-guided grasping, placement and handoff, increasing full-task success rate from 31.4% to 77%.
- Developed reinforcement learning pipelines for robotic manipulation, exploring policy optimisation methods such as PPO and SAC for simulated grasping and object-handoff tasks, improving task execution speed by 46.2%.

**Autonomous Robot Path Planning and Control**, Beijing Jiaotong University Jun 2025 – Aug 2025

- Implemented sensor acquisition, finite-state-machine logic and PID motor control on an STM32, integrating and testing these functions with host-computer perception and navigation modules for autonomous route execution.
- Integrated colour, ultrasonic and other sensors, developing feedback-based decision and control logic for line following and obstacle avoidance.
- Tuned sensing thresholds and PID gains through repeated track tests, coordinating perception, planning and control for autonomous navigation and obstacle avoidance. Increased driving speed by 61.1% over the initial baseline and achieved a 93.1% obstacle-avoidance success rate.

**Thermoelectric Autonomous Vehicle Design**, Beijing Jiaotong University Nov 2024 – Apr 2025

- Designed a thermally powered autonomous vehicle using SolidWorks, developing cam and gear transmission mechanisms for thermal-to-mechanical energy conversion.
- Integrated line-following hardware and completed mechanical assembly, calibration and system-level debugging, improving vehicle stability and reliability and achieving an 87.3% task success rate.

**Reconfigurable Land–Air Amphibious UAV**, Beijing Jiaotong University Mar 2024 – Aug 2024

- Developed and debugged the UAV flight-control system, refining attitude stabilisation, takeoff and landing control, and PID parameters through repeated whole-platform testing.
- Implemented and debugged ground–flight mode-switching logic, coordinating DC hub motors, four independently actuated servos and brushless propulsion motors for safe and responsive transitions between terrestrial driving and aerial flight.

## WORK EXPERIENCE

**Software Engineer Intern —Embodied AI**, Anyverse Technology Co., Ltd. Aug 2026 – Present

- Refined C++ motion control and trajectory planning for an in-house 7-DoF robotic arm, implementing forward and inverse kinematics (FK/IK) and five standard paths: straight-line motion along the X, Y and Z axes, square paths and circular paths. Enabled storage and cyclic execution of up to 64 taught waypoints. In limited real-robot tests of these paths, the largest measured peak joint-tracking error was approximately 0.12°.
- Optimised mixed-precision inference and inference backends for robot policies, evaluating PyTorch, TensorRT and graph-optimisation configurations. In fixed-input offline experiments with four camera inputs and a 26-dimensional action output, reduced median inference-call latency, measured after preprocessing, from 255.51 ms to 175.27 ms—a 31.40% reduction—while jointly assessing action-output deviations, end-effector pose differences and GPU memory usage.

• Extended a humanoid robot's C++/Python control SDK, developing over 30 interfaces and writing drivers for encoder-state access, PVT trajectory execution, trajectory export, dynamics and controller diagnostics to support real-robot experiments.
• Designed and conducted experiments on continuous robot-camera sampling and image–joint-state pairing, completing 49 sampling tests totalling 673.10 camera-stream hours and analysing 24 sets of image and joint-state recordings. Offline evaluation of nearest-timestamp pairing across 78,825 valid image frames yielded median absolute time differences of 2.43–2.59 ms, providing a quantitative basis for constructing visual and proprioceptive observations.
• Designed and implemented an automated evaluation and iteration loop for the 100 BEHAVIOR Challenge tasks, connecting task execution, AI-based motion detection, result evaluation and feedback-driven refinement. Reduced review time by 60.1%; the team ultimately achieved world-leading results in the competition.

**Human Biomechanics & Embodied Simulation Intern**, Chongqing Xingshijie Co., Ltd. Jan 2026 – Mar 2026
• Built human musculoskeletal models in OpenSim and imported human-motion datasets, analysing over 100 model data sources covering 18 types of human movement to establish a simulation workflow for whole-body motion and joint biomechanics.
• Computed and analysed joint forces, loads and kinematic profiles across 156 motion sequences and 23 major joints, identifying high-load phases and biomechanical patterns relevant to movement assessment and injury prevention.
• Translated simulation results into human-motion and biomechanical priors for embodied AI, providing joint-level references for human motion understanding, humanoid motion representation and human-like robot control.

**Robot Learning and Manipulation Intern**, Csoft Jul 2024 – Aug 2024
• Built a ROS/Ubuntu ALOHA bimanual robotic-arm system for teleoperation-based human-action imitation, supporting object grasping, transfer, book handling and water pouring, with a 77.3% task success rate across 220 real-robot trials.
• Integrated dual-arm control, teleoperation, joint-state feedback, calibration and demonstration execution into an end-to-end manipulation pipeline, completing 36 sets of calibration and demonstration-execution tests to reproduce human-operated behaviours.
• Diagnosed hardware–software synchronisation and calibration errors through repeated real-robot trials, improving execution consistency for contact-rich grasping, transfer and pouring tasks and reducing the number of task failures during repeated execution by 28.6%.

## AWARDS

- Second Prize, Beijing College Students Engineering Practice and Innovation Competition Mar 2025
- Third-Class Academic Excellence Scholarship 2024-2025, 2025-2026
- Third-Class Scholarship for Outstanding Social Work 2024-2025, 2025-2026
- Second-Class Sports Activity Scholarship 2023–2024

## LEADERSHIP & EXTRACURRICULAR ACTIVITIES

**Captain,** Basketball Team, School of Mechanical Engineering Jun 2024 – Present
- Managed team operations, scheduling, and communication among players and coaching staff.
- Led the team to a top-four finish in the university basketball tournament.

**Deputy Director,** Visual Arts Department, University Arts Center Oct 2024 - Oct 2025
- Coordinated campus visual-media projects and produced photography/video content for university events.

## SKILLS AND INTERESTS

**Programming Languages:** Python, C++, C, MATLAB
**Robotics:** ROS/ROS 2, Robot Manipulation, Motion Planning, Motion Control, Sim-to-Real
**AI / ML:** PyTorch, Deep Learning, Robot Learning, Computer Vision
**Tools:** Linux, Git, MATLAB/Simulink, MuJoCo / Isaac Sim
**Languages:** Chinese (Native), English (IELTS: 7.5)
**Interests:** Basketball (University Team Captain), Photography, Video Production, Reading, Robotics, Artificial Intelligence